



Degradation of 1,2,3-trichloropropane by pyrite activating sodium percarbonate and the implications for groundwater remediation

Chunyun Gu, Shuyu Liu, Jiarui Liang, Yuqian Wang, Shuyu Lu, Jie Ma^{*,1}

State Key Laboratory of Heavy Oil Processing, Beijing Key Lab of Oil & Gas Pollution Control, China University of Petroleum-Beijing, Beijing 102249, China

ARTICLE INFO

Keywords:

In-situ chemical oxidation
Contaminated site
Groundwater remediation
Chlorinated hydrocarbons
Fenton like
Percarbonate

ABSTRACT

1,2,3-trichloropropane (TCP) is a newly regulated groundwater contaminant, however, its remediation technology has not been fully investigated. In this research, pyrite (FeS_2) was used as a novel activator of sodium percarbonate (SPC) for TCP degradation. TCP could be effectively removed by FeS_2 /SPC system with a much higher removal efficiency than the classic Fenton system. Free radical quenching test, hydroxyl radical probe reaction, EPR, XPS, XRD and SEM were used to elucidate the reaction mechanism. SPC was decomposed to produce H_2O_2 , which was activated by Fe^{2+} dissolved from pyrite to produce $\text{HO}\cdot$ and $\text{O}_2\cdot^-$. Both $\text{HO}\cdot$ and $\text{O}_2\cdot^-$ contributed to TCP degradation, but $\text{HO}\cdot$ played a more important role. The S elements of pyrite promoted the regeneration of Fe^{2+} , thus enhancing the TCP degradation. The TCP removal efficiency significantly increased with increases in concentrations of pyrite and SPC and decreases in pyrite particle size and pH. Although HCO_3^- , Cl^- and NOM inhibited TCP degradation by FeS_2 /SPC, the reaction system could still remove TCP in real groundwater. Overall, pyrite-activate SPC is a promising in-situ groundwater remediation technology.

1. Introduction

1,2,3-trichloropropane (TCP) is a byproduct of epichlorohydrin synthesis and used as a solvent to clean and degrease operations [1–3]. As an emerging groundwater contaminant, it has attracted great attention from regulators of Europe [4] and US [5,6]. U.S. Environmental Protection Agency (USEPA) added TCP to its Unregulated Contaminant Monitoring Rule (UCMR) 3 with minimum reporting level of 0.03 $\mu\text{g/L}$ [7]. Many US states established the health-based drinking water guidance, including the California (0.0007 $\mu\text{g/L}$), Colorado (0.00037 $\mu\text{g/L}$), Minnesota (0.003 $\mu\text{g/L}$), New Jersey (0.013 $\mu\text{g/L}$) and Texas (0.03 $\mu\text{g/L}$) [8], which are much lower than the maximum contaminant level for TCE (trichloroethylene) or PCE (tetrachloroethene) (5 $\mu\text{g/L}$). This compound has been detected in hundreds of surface water and drinking water sources at levels of 0.1–100 $\mu\text{g/L}$ in the US [9] and up to 9 $\mu\text{g/L}$ in groundwater of the Netherlands [10]. Up to 28.76 mg/L of TCP was detected in a contaminated aquifer of China [11]. This compound is resistant to both abiotic and biotic degradation pathways making it persistent in the environment [12]. Several technologies have recently investigated for TCP removal, including granular activated carbon [8], anaerobic bioremediation [13] and chemical reduction (e.g., zero-valent

metals) [14]. However, applicability of chemical oxidation technique has not been investigated.

In-situ chemical oxidation (ISCO) is a widely used in-situ groundwater remediation technology. The Fenton process is one of the most popular ISCO techniques [15–19]. However, for in-situ applications, the rapid precipitation of $\text{Fe}(\text{OH})_3$ has already been known to be major limitations of Fenton reactions [20]. In addition, H_2O_2 is unstable and has a relatively short lifetime and radius of influence after injection into the subsurface [21–23]. Therefore, various heterogeneous Fenton systems have been developed, which use solid catalyst to replace soluble Fe^{2+} or solid oxidant as the source of H_2O_2 to overcome the limitations of classic Fenton technique. In the last 5–10 years, heterogeneous Fenton systems that use pyrite instead of soluble Fe^{2+} have been developed [20, 23–33]. Pyrite is one of the most abundant natural iron sulfur minerals on the earth and showed high catalytic activities. For example, Che et al. [20] found that the pyrite/ H_2O_2 system effectively degraded trichloroethylene. Zhou et al. [23] used pyrite to activate calcium peroxide to efficiently transform diethyl phthalate. Kim et al. [27] found that 0.1 mM sulfanilamide was oxidized to 80% by pyrite/ CaO_2 . The use of natural pyrite avoids injection of external activators into the aquifer, thus reducing environmental disturbance.

* Corresponding author.

E-mail address: rubpmj@sina.com (J. Ma).

¹ ORCID 0000-0003-3719-3814

Sodium percarbonate ($2\text{Na}_2\text{CO}_3 \cdot 3\text{H}_2\text{O}_2$, SPC) is a promising solid oxidant for practical applications [21,27,34–36]. Compared with H_2O_2 , SPC is more stable and safer to handle and transport [37]. In previous studies, the heterogeneous Fenton system could degrade the organic contaminants by generating highly reactive species from the reaction of $\text{Fe}^{2+}/\text{Fe}^{3+}$ with SPC [18,19,37,38]. However, there has not been any report on the use of iron bearing minerals for SPC activation to degrade organic contaminants.

This research aimed to evaluate the feasibility of applying pyrite-activated SPC for TCP removal. X-ray photoelectron spectroscopy (XPS), X-ray diffraction (XRD), scanning electron microscopy (SEM), free radical quenching test, electron paramagnetic resonance (EPR) detection and hydroxyl radical probe reaction were used to explore the reaction mechanism. The mineralization and dechlorination of TCP were elucidated through the measurement of total organic carbon (TOC) and the released Cl^- . The influence of SPC concentration, pyrite concentration, pyrite particle size and initial pH on TCP degradation were investigated. The effects of solution matrix on TCP degradation were also explored. Application of pyrite-activated SPC to remove TCP in real groundwater was also investigated.

2. Materials and methods

2.1. Materials

Pyrite (FeS_2) was collected at Shangbao City in Hunan Province, China. Sodium percarbonate ($2\text{Na}_2\text{CO}_3 \cdot 3\text{H}_2\text{O}_2$, $\geq 13.0\%$ active oxygen), hydrogen peroxide (H_2O_2 , 30% wt), tert-butyl alcohol (TBA, $\geq 99.5\%$), methanol (MeOH, 99.5%), p-Benzoquinone (p-BQ, 99%), chloroform (CHCl_3 , CF, 99%), 1,10-phenanthroline (99%), hydroxylammonium chloride (98.5%), sulfuric acid (H_2SO_4 , 98%), sodium hydroxide (NaOH, 97%), sodium chloride (NaCl, 99.5%), sodium sulfate anhydrous (Na_2SO_4 , 99%), sodium bicarbonate (NaHCO_3 , $\geq 99.8\%$), sodium nitrate (NaNO_3 , 99%), p-Hydroxybenzoic acid (p-HBA, 99%) and benzoic acid (BA, 99.5%) were obtained from Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). 1,2,3-Trichloropropane (TCP, 98%) was purchased from Adamas Reagent, Ltd. (Shanghai, China). The TCP in excess was added into 18.2 mΩ high-purity water to form a TCP saturated solution (11.87 mM), which was used as the stock solution of TCP [39]. Humic acid (HA) was provided by Sinopharm Chemical Reagent Co., Ltd. (Beijing, China). The pyrite was crushed with an agate mortar and separated with 200 mesh. Pyrite diffractograms agreed well with pyrite diffraction peaks in Joint Committee on Powder Diffraction Standards (JCPDS) diffraction data files (JADE Materials Data Inc. card No. 71–2219) (Fig. S1) [40]. The surface morphology of pyrite was visualized by SEM micrography. The SEM image indicated that pyrite was irregular and cubic particles (Fig. S2). The real groundwater samples were collected directly from the wellhead of wells in sterile glass bottles. Until further analysis or processing, all samples were stored at 4 °C in the laboratory [41]. The sampling locations and anion concentrations of real groundwater samples were provided in Table S1.

2.2. Experimental procedure

Batch experiments were carried out in 50-mL amber-colored glass vials containing 25 mL reaction solution. The glass vials were capped with Teflon Mininert valves and incubated on an orbital shaker (HYG-B) at 25 °C in the dark [42]. The reaction solution contained 0.1 mM TCP and 1 mM SPC. The initial pH was adjusted to 3 by H_2SO_4 . No pH buffer was added, since it may interfere with the reaction [20,23]. FeS_2 (5 g/L) was then added to initiate the reaction. The 200 μL headspace gas was taken to monitor TCP. Classic Fenton experiment were carried out as a control using the 1 mM Fe^{2+} /1 mM H_2O_2 and 41.7 mM Fe^{2+} /1 mM H_2O_2 . The 5 g/L FeS_2 was equivalent to 41.7 mM Fe^{2+} . All experiments were conducted in triplicate.

2.3. Analysis

Pyrite particle was analyzed by Gemini SEM 300 (ZEISS) scanning electron microscopy (SEM). X-ray diffraction (XRD) measurements were executed using an Empyrean instrument (PANalytical) with a Cu K α radiation source at 40 kV and 40 mA. X-ray photoelectron spectroscopy (XPS) analysis was recorded on a K-Alpha instrument (Thermo) with monochromatized Al K α radiation. Electron paramagnetic resonance (EPR) technology was employed to confirm the generation of free radicals in the FeS_2 /SPC system. The concentrations of TCP in headspace were converted to those in aqueous solution by using dimensionless Henry's law constants [43]. The TCP was quantified by an Agilent 7890B gas chromatography (GC) equipped with a HP-5 column (30 m $\times$ 0.32 mm, 0.25 μm) and an electron capture detector (ECD). The Injector temperature was 220 °C. The detector temperature was 280 °C. The oven temperature was 100 °C. The intermediates of TCP were identified by a GC/MS (7890B/5977B) with HP-5MS column (30 m $\times$ 0.25 mm, 0.25 μm) using National Standard in Chinese HJ 639–2012.

The hydroxylation of BA to p-HBA as a hydroxyl radical probe reaction was used to quantify the amount of hydroxyl radicals [44]. The p-HBA concentrations were measured using HPLC equipped with an Eclipse Plus C18 column (4.6 $\times$ 250 mm, 5 μm) and a UV detector (255 nm). The flow rate was 1 mL/min. The mobile phase consisted of 0.1% trifluoroacetic acid and acetonitrile (65:35, v/v). A conversion factor of 5.87 was used to estimate the cumulative $\text{HO}\cdot$ concentrations.

Cl^- and SO_4^{2-} produced from the degradation reaction were determined using an ion chromatograph (IC) (Dionex ICS-1100) equipped with an IonPac AS22 analytical column (Dionex 4 $\times$ 250 mm) [45]. Dissolved ferrous ions and ferric ions were quantified by the 1,10-phenanthroline method with the UV–vis spectrophotometer at a wavelength of 510 nm. The concentration of H_2O_2 produced from the dissolution of SPC was confirmed by the potassium titanium oxalate spectrophotometric method with the UV–vis spectrophotometer at a wavelength of 402 nm. Total organic carbon (TOC) was measured by TOC-L (Shimadzu). The concentration of HA was expressed as dissolved organic carbon measured by TOC-L (Shimadzu) [46].

3. Results and discussion

3.1. TCP degradation by FeS_2 /SPC

FeS_2 /SPC system could remove TCP effectively with a much higher efficiency than the classic Fenton system (Fig. 1). 91.3% of TCP (0.1 mM) was removed within 240 min by 5 g/L pyrite coupled with 1 mM

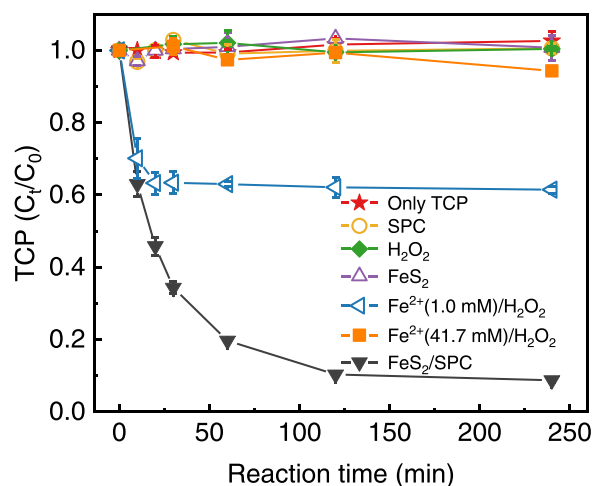
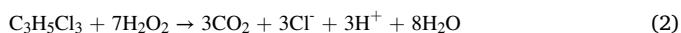
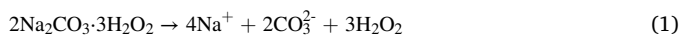


Fig. 1. TCP degradation in various systems. Experimental condition: $[\text{TCP}]_0 = 0.1 \text{ mM}$; $[\text{SPC}]_0 = 1 \text{ mM}$; $[\text{H}_2\text{O}_2]_0 = 1 \text{ mM}$; $[\text{FeS}_2]_0 = 5 \text{ g/L}$; $[\text{Fe}^{2+}]_0 = 1 \text{ mM}$ or 41.7 mM; initial pH = 3.

SPC. In contrast, 38.5% of TCP was removed by 1 mM Fe²⁺ coupled with 1 mM H₂O₂ and only 5.7% of TCP was removed by 41.7 mM Fe²⁺ coupled with 1 mM H₂O₂. Because S elements of pyrite reduced Fe³⁺ to Fe²⁺, thus promoting the regeneration of Fe²⁺ and enhancing TCP removal. More details were discussed in Section 3.3.

The dechlorination and mineralization of TCP were elucidated through the measurement of the released Cl⁻ and TOC. Fig. 2 showed that about 0.0913 mM TCP was removed by FeS₂/SPC system. Theoretically, the complete dichlorination of 0.0913 mM TCP would release 0.274 mM Cl⁻ (Eq. (2)). The measured Cl⁻ concentration was 0.283 mM. Therefore, TCP was completely dechlorinated within 240 min. In addition, Fig. 2 showed that TOC decreased with the TCP removal and the residual TOC decreased to 51.1% within 240 min. Therefore, TCP was not completely mineralized, leaving some organic intermediates in the solution. However, no organic intermediates could be detected by GC/MS (Fig. S3). It might be due to the low concentration of organic intermediates. The results demonstrated that FeS₂/SPC system was effective in the degradation of TCP but the reaction pathways of TCP needed to be further studied.



3.2. Free radical generated in the FeS₂/SPC system

To characterize the produced free radicals, TBA, CF and p-BQ were used as radical scavengers. TBA was employed as HO· scavenger ($k_{\text{TBA}, \text{HO}\cdot} = 3.8\text{--}7.6 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$). CF scavenged O₂· effectively ($k_{\text{CF}, \text{O}_2\cdot} = 3 \times 10^{10} \text{ M}^{-1}\text{s}^{-1}$) but hardly reacted with HO· ($k_{\text{CF}, \text{HO}\cdot} = 7 \times 10^6 \text{ M}^{-1}\text{s}^{-1}$). p-BQ effectively scavenged both O₂· ($k_{\text{p-BQ}, \text{O}_2\cdot} = 9.6 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$) and HO· ($k_{\text{p-BQ}, \text{HO}\cdot} = 1.2 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$) [37,47–51]. When the TBA concentration increased from 0 to 10 mM, the TCP removal efficiency decreased from 89.2% to 13.5% (Fig. 3(a)), indicating that HO· played an important role (Eq. (4)). However, 13.5% of TCP could still be removed even in the presence of 10 mM TBA, indicating that a non-HO· degradation mechanism existed. The TCP removal efficiency decreased from 89.2% to 21.8% respectively with the CF concentrations increased from 1 to 10 mM (Fig. 3(b)), suggesting that O₂· also participated in the reaction (Eqs.(5–7)). The TCP removal was completely inhibited when the p-BQ concentration reached 10 mM (Fig. 3(c)), corroborating that both HO· and O₂· contributed to TCP degradation [34,52]. In summary, free radical quenching experiments showed that: (1) FeS₂/SPC system produced HO· and O₂·; (2) both HO· and O₂· contributed to TCP degradation.

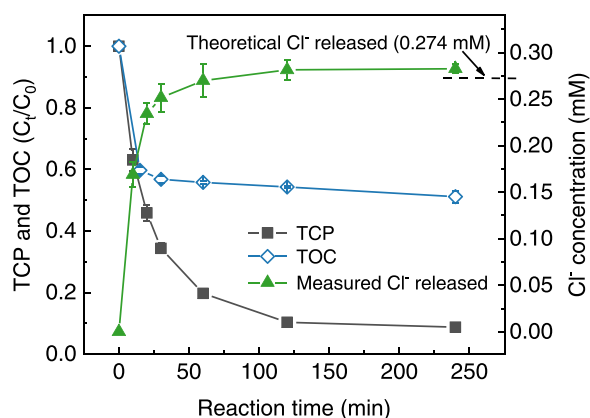
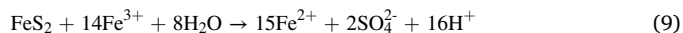
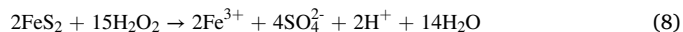
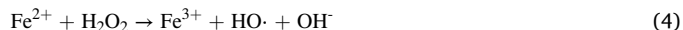


Fig. 2. The changes of TCP, TOC and the generation of Cl⁻ with reaction time in FeS₂/SPC system. Experimental condition: [TCP]₀ = 0.1 mM; [SPC]₀ = 1 mM; [FeS₂]₀ = 5 g/L; initial pH = 3.



In order to further clarify the presence of the free radicals, the EPR spectrum was used to detect HO· and O₂· in the FeS₂/SPC system. 5,5-dimethyl-1-pyrrolidine-N-oxide (DMPO) were applied as the spin capture reagents. Four peaks with an intensity ratio of 1:2:2:1 represented the spin adduct DMPO-HO· (Fig. 4(a)). Obviously, HO· produced in the FeS₂/SPC system and the HO· abundance gradually increased with time [53]. Fig. 4(b) showed that the six typical peaks of DMPO-O₂·, suggesting the FeS₂/SPC system produced O₂· and the O₂· abundance also increased with time [54]. In addition, HO· had the higher intensity than O₂· (Fig. 4), suggesting that HO· played a more important role than O₂·.

The hydroxylation of BA to p-HBA as a hydroxyl radical probe reaction was used to quantify the HO· concentration [44]. Fig. 5(a) showed that the cumulative concentration of HO· produced within 240 min was 0.235 mM and HO· production was most prominent during the first 10 min which was consistent with the trend of TCP degradation (Fig. 1). Theoretically, 1 mM SPC contains 1.5 mM H₂O₂ while the measured HO· concentration was only 0.235 mM. This is because HO· would be scavenged according to Eqs. (10–13) [21,34,55]. Miao et al. [34] reported 0.15 mM of HO· in Fe(II)/SPC/hydroxylamine hydrochloride system which contained 0.5 mM SPC. Sun et al. [56] reported 0.264 mM HO· was generated in siderite (1 g/L)/H₂O₂ (10 mM) system. Kim et al. [27] reported 0.15 mM HO· was generated in pyrite (50 g/L)/CaO₂ (1 wt%) system. In contrast to the FeS₂/SPC system, HO· was not detected in the SPC only system while only a small amount of HO· (0.006 mM) was detected in the FeS₂ only system (Fig. 5(b)). The FeS₂ only solution contained dissolved oxygen. Previous studies have reported that FeS₂ could react with dissolved oxygen to produce HO· [39,44,57], therefore a small amount of HO· was generated in the FeS₂ only system.



3.3. Fe²⁺ regeneration and pyrite characterization

Temporal changes in Fe²⁺, Fe³⁺ and SO₄²⁻ concentration as well as XPS characterization of pyrite indicated that S elements of pyrite acted as an electron donor to reduce Fe³⁺ to Fe²⁺, thus promoting the regeneration of dissolved Fe²⁺ and the redox cycle of dissolved Fe³⁺/Fe²⁺. The regeneration of Fe²⁺ in FeS₂/SPC system overcame the drawback that dissolved Fe²⁺ could not be regenerated in classic Fenton system, thus promoting the HO· formation and the contaminant degradation. This explained why TCP degradation by FeS₂/SPC was much faster than Fe²⁺/SPC (Fig. 1). Detailed discussion on Fe²⁺ regeneration was as follows.

Fe²⁺ was leached from the pyrite (Eq. (3)). Once pyrite was added, a quick increase of the total dissolved iron concentration was observed and the total dissolved iron was composed mainly of Fe²⁺ before SPC

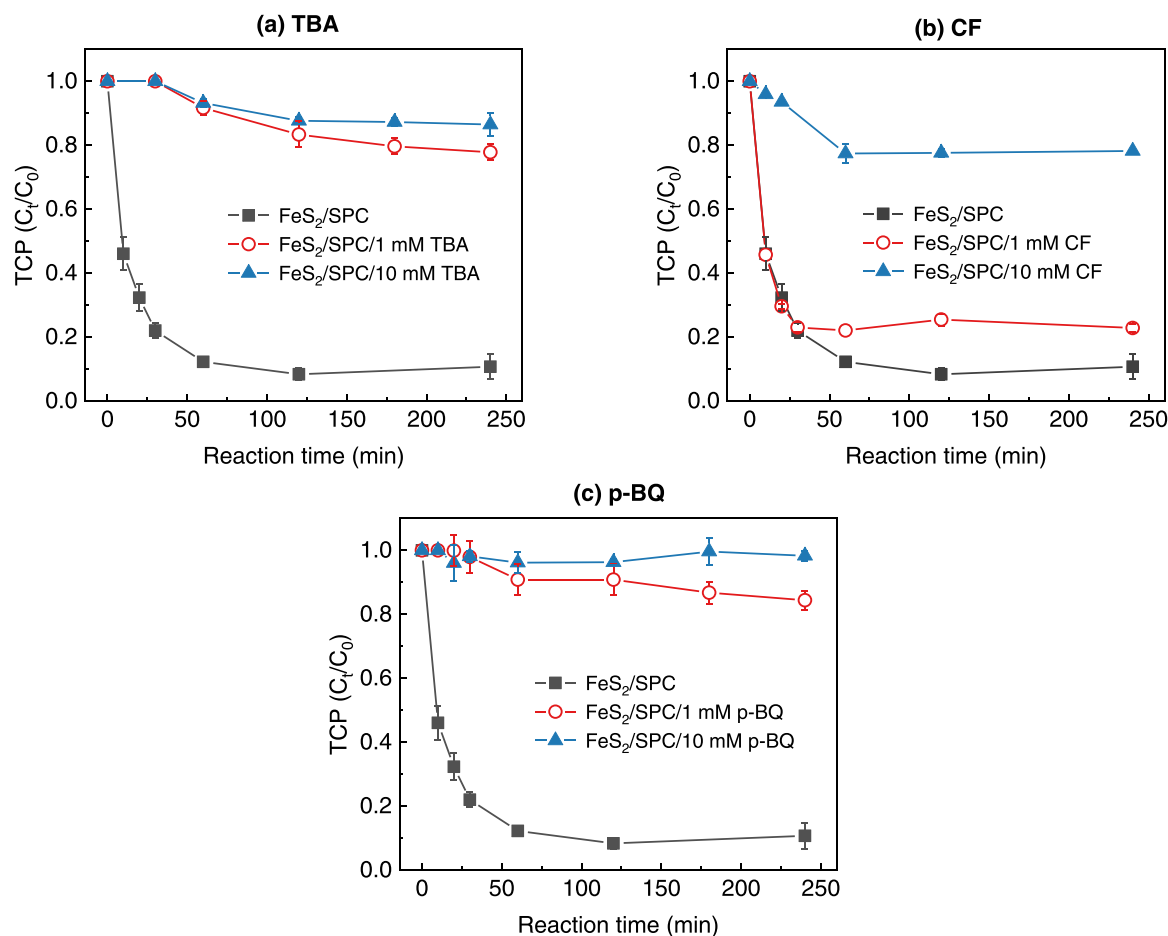


Fig. 3. Effect of different radical scavengers on TCP degradation: (a) TBA; (b) CF; (c) p-BQ. Experimental condition: $[TCP]_0 = 0.1$ mM; $[SPC]_0 = 1$ mM; $[FeS_2]_0 = 5$ g/L; initial pH = 3.

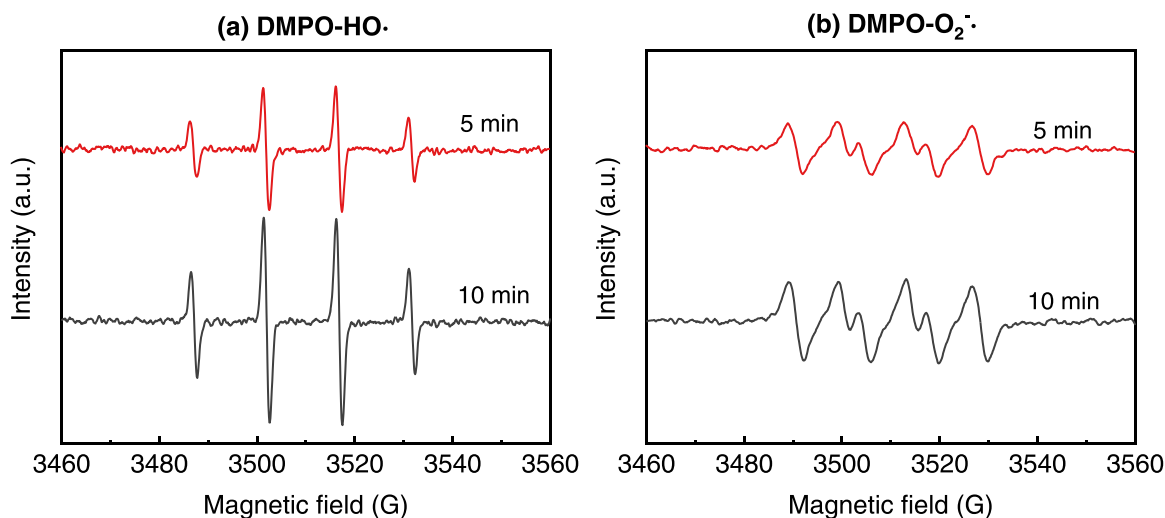


Fig. 4. EPR spectra of the FeS₂/SPC system: (a) DMPO-HO·; (b) DMPO-O₂^{·-}. Experimental condition: $[TCP]_0 = 0.1$ mM; $[SPC]_0 = 1$ mM; $[FeS_2]_0 = 5$ g/L; initial pH = 3.

addition (Fig. 6). After SPC addition (from 0 to 240 min), The total dissolved iron concentration kept increasing because the SPC-released H₂O₂ reacted with FeS₂ to produce Fe³⁺ (Eqs. (4) and (8)). However, the total dissolved iron concentration reached a plateau after 240 min (Fig. 6), suggesting that there was not much H₂O₂ remained to react with FeS₂. Changes in Fe²⁺ and Fe³⁺ concentrations were more complicated

(Fig. 6). From 0–30 min, Fe²⁺ concentration decreased significantly while Fe³⁺ concentration increased. After 30 min, Fe²⁺ increased while Fe³⁺ decreased. Increases in Fe²⁺ concentration were due to the reduction of Fe³⁺ to Fe²⁺ by S element of pyrite (Eq. (9)), which promoted the regeneration of Fe²⁺. In addition, the SO₄²⁻ concentration in the FeS₂/SPC system was almost twice of that in the FeS₂ only system

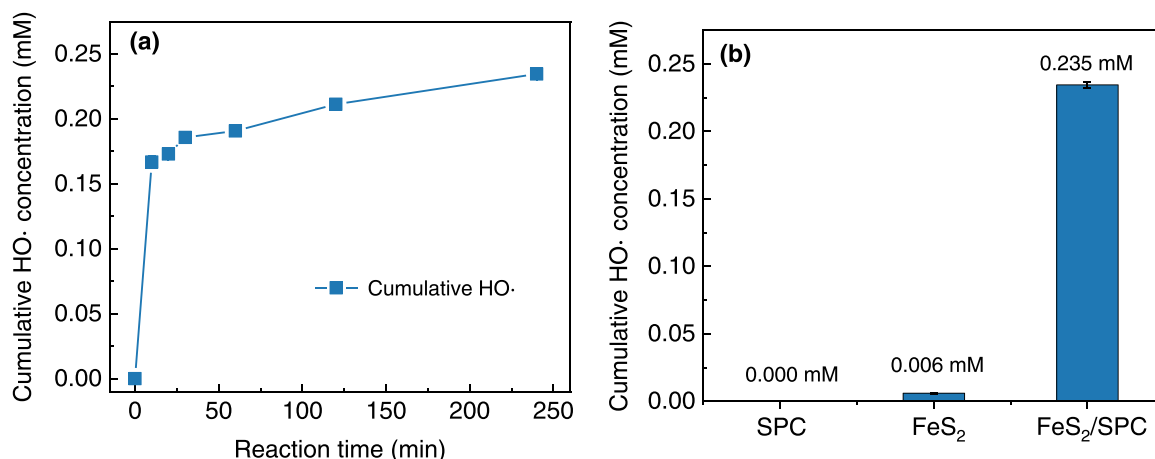


Fig. 5. Cumulative amounts of hydroxyl radicals: (a) Temporal changes in the concentration of cumulative hydroxyl radicals; (b) cumulative amounts of hydroxyl radicals within 240 min in different reaction systems. Experimental condition: [BA]₀ = 1 mM; [SPC]₀ = 1 mM; [FeS₂]₀ = 5 g/L; initial pH = 3.

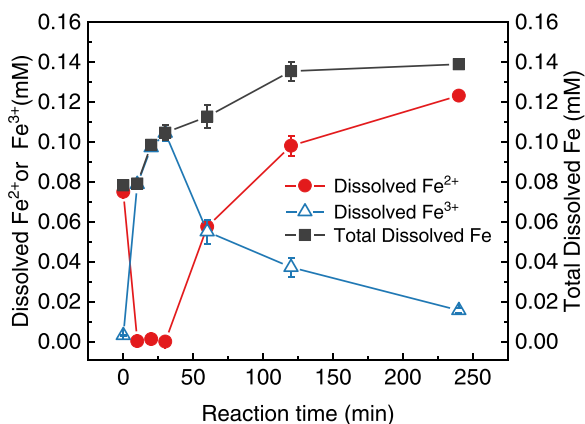


Fig. 6. Temporal changes in the concentration of Fe²⁺, Fe³⁺ and total dissolved Fe in the FeS₂/SPC system. Experimental condition: [SPC]₀ = 1 mM; [FeS₂]₀ = 5 g/L; initial pH = 3. The concentration at time 0 was measured without adding SPC.

(Fig. S4), because the S elements of pyrite were oxidized to SO₄²⁻ in the process of reducing Fe³⁺ to Fe²⁺. This phenomenon also confirmed that S elements of pyrite participated in Fe²⁺ regeneration. Zhou et al. [23] also found that the S elements of pyrite promoted the regeneration of Fe (II).

The XPS analysis of pyrite was showed in Fig. S5(a-d). The peak at 708.1 eV and 710.8 eV represented the binding energy of Fe(II) and Fe (III), respectively. Meanwhile, the peak at 163.3 eV, 164.5 eV and 169.7 eV represented the binding energy of S₂²⁻, S_n²⁻ and SO₄²⁻, respectively. XPS analysis showed that the ratio of Fe(II) increased from 72.4% to 74.3% and the proportion of Fe(III) decreased from 27.6% to 25.7% within 240 min. Increases in Fe(II) content and decreases in Fe(III) content corroborated that Fe(III) could be reduced to Fe(II) by the S elements of pyrite. The XPS spectra of S2p_{3/2} showed that: (1) the proportion of S₂²⁻ decreased from 62.9% to 62.4%; (2) the proportion of S_n²⁻ decreased from 27.5% to 25.9%; (3) the proportion of SO₄²⁻ increased from 9.6% to 11.7%. This indicated that the S elements of pyrite were oxidized during the reaction, which further proved that S elements of pyrite reduced Fe³⁺ to Fe²⁺ and promoted the Fe²⁺ regeneration.

3.4. Summary of reaction mechanism

Based on the results of Sections 3.1, 3.2 and 3.3, the mechanism of TCP degradation by FeS₂/SPC was summarized as follows (Fig. S6): (1)

SPC dissolution produced H₂O₂; (2) Fe²⁺ released from pyrite activated H₂O₂ to form O₂^{·-} and HO·; (3) both HO· and O₂^{·-} contributed to TCP oxidation, but HO· played a more important role; (4) the S elements of pyrite reduced Fe³⁺ to Fe²⁺, thus promoting the regeneration of Fe²⁺ and further production of HO· and O₂^{·-}.

3.5. Effects of SPC concentration, pyrite concentration, pyrite particle size and initial pH on TCP degradation

The removal efficiency of TCP increased from 18.4% to 96.3% when the SPC concentration increased from 0.1 mM to 5 mM (Fig. 7(a)). Increases in concentration from 0.1 to 5 mM led to proportional increases in H₂O₂ concentration from 0.15 to 7.5 mM (No TCP was added.) (Fig. S7). More H₂O₂ resulted in more HO· formed and thus more TCP removed. However, when the SPC concentration increased from 5 mM to 20 mM, the TCP removal efficiency reduced from 96.3% to 35% (Fig. 7(a)). Similar results have been reported by Chen et al. [45] for the degradation of 2,4-dichlorophenoxyacetic acid using FeS/ H₂O₂. When the SPC concentration further increased from 5 to 20 mM, the H₂O₂ concentration increased from 7.5 mM to 28.8 mM (Fig. S7). Excess H₂O₂ reacted with HO· to form hydroperoxide radicals (HO₂[·]) that were less reactive (Eqs.(12–13)). Therefore, the TCP removal efficiency decreased when the SPC concentration was higher than 5 mM [55].

Fig. 7(b) showed that the TCP removal efficiency increased with the increasing FeS₂ concentration. Higher pyrite concentration resulted in more Fe²⁺ released and TCP removed (Fig. S8). However, when the FeS₂ concentration further increased from 5 g/L to 10 g/L, the TCP removal efficiency had only minimal increase from 86.8% to 89.2%, which was probably because excessive Fe²⁺ react with HO· (Eq. (11)) resulting in HO· scavenging [21,45]. Yan et al. [58] also found that degradation of TCE was inhibited when siderite concentration exceeded a threshold value (0.1145 g) in the siderite/Na₂S₂O₈/H₂O₂ system.

Fig. 7(c) showed that the TCP removal efficiency increased with the decreasing pyrite particle size. Decrease in pyrite particle size led to increase in the specific surface area, thus resulting in more Fe²⁺ released. Fig. S9 showed that when pyrite particle size decreased from 20 mesh to 200 mesh, the Fe²⁺ concentration increased from 0.01 mM to 0.09 mM. Higher Fe²⁺ concentration resulted in more SPC activated and thus higher TCP removal efficiency. Zhang et al. [59] also found that decreasing pyrite particle size resulted in increasing Fe²⁺ concentration and p-chloroaniline removal efficiency.

Fig. 7(d) showed that TCP could be effectively removed at initial pH of 3–9 and the removal efficiency increased with the decreasing initial pH. The increasing pH led to Fe²⁺ precipitation, which inhibited SPC activation and HO· formation. The cumulative amounts of HO·

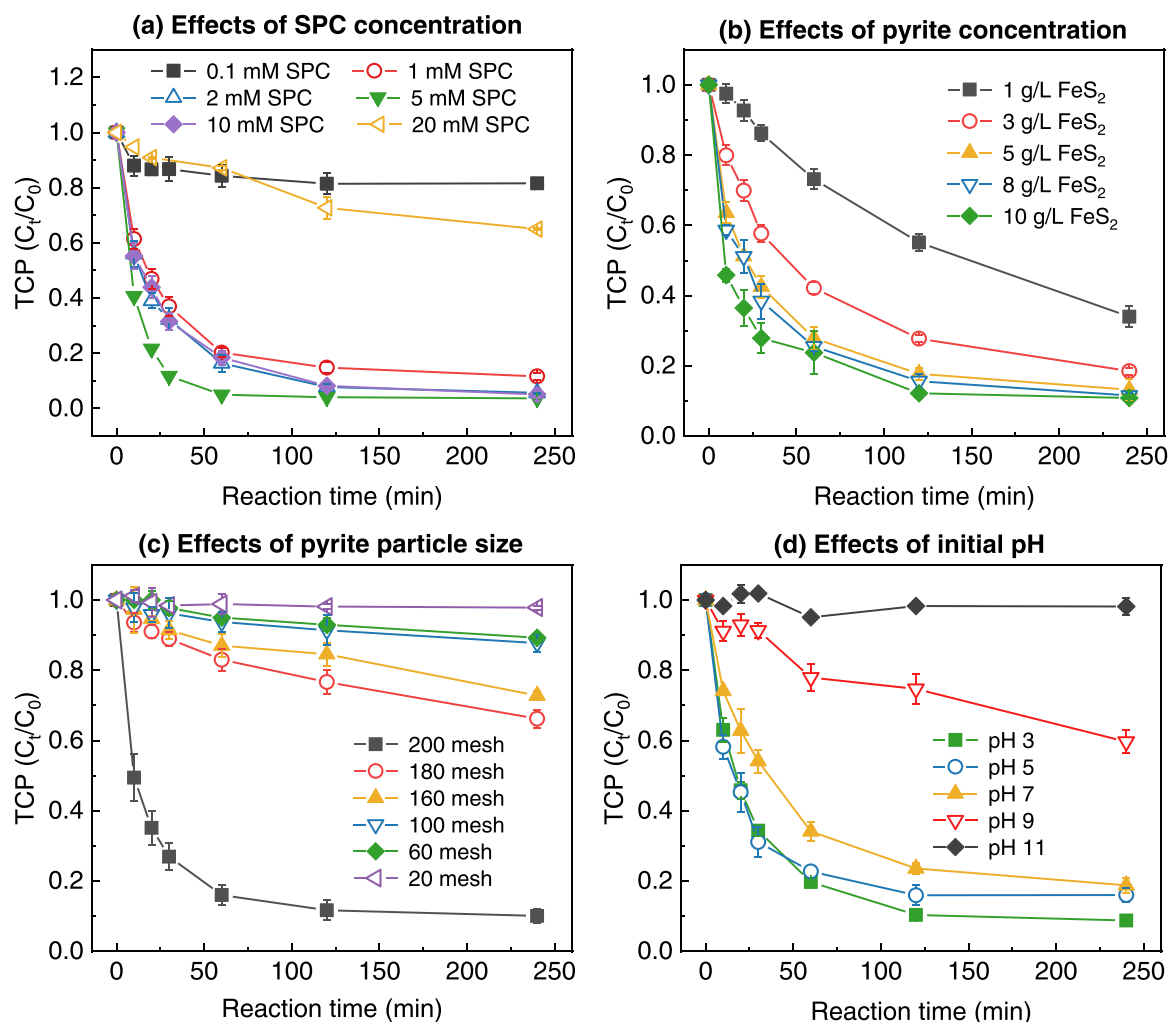


Fig. 7. Effect of SPC, pyrite and initial pH on TCP degradation: (a) Effects of SPC concentration; (b) effects of pyrite concentration; (c) effects of pyrite particle size; (d) effects of initial pH. Experimental condition: [TCP]₀ = 0.1 mM; [SPC]₀ = 1 mM; [FeS₂]₀ = 5 g/L; initial pH = 3.

decreased from 0.235 mM to 0.005 mM when the initial pH increased from 3 to 11 (Fig. S10). Therefore, TCP removal efficiency decreased with the increasing initial pH. In addition, Fig. S11 showed that the final pH was lower than the initial pH in all treatments. TCP oxidation produced H⁺, thus resulting in decreasing pH (Eq. (2)).

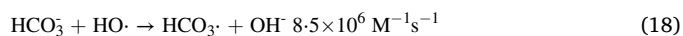
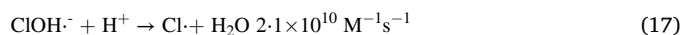
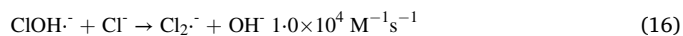
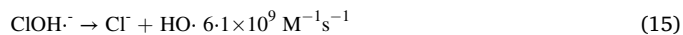
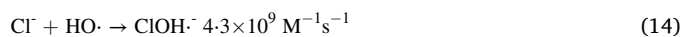
3.6. Effects of environmental anions and NOM on TCP degradation

HCO₃⁻, Cl⁻ and NOM inhibited FeS₂/SPC oxidation of TCP and the inhibitory effects increased with increasing concentrations of HCO₃⁻, Cl⁻ and NOM. HCO₃⁻ had stronger inhibitory influence than Cl⁻ (Fig. S12(a) and (b)). The reaction was significantly inhibited when the NOM concentration was higher than 1 mg/L (Fig. S13). In contrast, SO₄²⁻ and NO₃⁻ had negligible impacts on TCP degradation.

The HCO₃⁻ and Cl⁻ inhibited the degradation of TCP, because HCO₃⁻ and Cl⁻ quenched HO· (Eqs.(14–18)) [60]. The quenching reaction generated weak oxidizing species, thus TCP degradation was inhibited. Theoretically, the inhibitory effect of Cl⁻ on TCP degradation was limited because the rate of Cl⁻ reacting with HO· and the rate of HO· regeneration was comparable (Eqs.(14–15)). However, the ClOH·⁻ could further react with Cl⁻ and H⁺ (Eqs.(16–17)), which hampered HO· generation via Eq. (15). In addition, the presence of HCO₃⁻ reacted with Fe²⁺ to form precipitation on the pyrite surface, which inhibited SPC activated by pyrite and TCP degradation. Therefore, HCO₃⁻ had stronger inhibitory influence than Cl⁻. Furthermore, NOM scavenged HO· thus inhibiting

TCP degradation. Wang et al. [61] found that humic acid not only consumed HO· and inhibited the degradation of the target compound, but also inhibited the regeneration of Fe(II).

In real groundwater, the maximum concentration of HCO₃⁻ was no more than 4.1 mM [62]. According to the results of Fig. S12(a), HCO₃⁻ would inhibit TCP degradation by FeS₂/SPC system to a certain extent in real water. Meanwhile, Cl⁻ concentration in low mineralized water was no more than 3 mM [62], which had little inhibition on TCP degradation and the FeS₂/SPC system could effectively remove TCP according to the results of Fig. S12(b). However, Cl⁻ concentration could reach 282 mM in high mineralized water [62]. Therefore, TCP degradation by FeS₂/SPC system would be severely inhibited in high mineralized water. Furthermore, in the natural environment, the target contaminant often coexisted with humic substance. The humic substance concentration in groundwater was the lowest (0.03–0.6 mg/L) [63]. According to the results of Fig. S13, the FeS₂/SPC system could effectively remove TCP in real groundwater.



3.7. Performance of FeS₂/SPC system in real groundwater

Feasibility and applicability of FeS₂/SPC system were assessed in real groundwater for TCP degradation (Fig. 8). Table S1 presented the main characteristics of four real groundwater samples. The TCP removal efficiencies in real groundwater samples were lower than in pure water, because the presence of matrix species in real groundwater (Table S1). As discussed in Section 3.6, HCO₃⁻, Cl⁻ and NOM consumed HO· and significantly inhibited TCP oxidation. In addition, TCP removal efficiency in groundwater #4 was lower than that in the other three groundwater samples. Because both TOC and Cl⁻ concentration of groundwater #4 were higher than that of the other three groundwaters. Although the TCP removal efficiency in real groundwater decreased, TCP could still be effectively removed. In summary, the pyrite-activated SPC is a promising technology for the remediation of TCP-contaminated groundwater. However, the inhibitory effect of solution matrix (e.g., HCO₃⁻, Cl⁻ and NOM) on TCP degradation should be taken in account in practical applications. In addition, the current technique was applied with pH adjustment (to 3) which is not a perfect idea for practical applications. Therefore, we this technique needs further improvement to accommodate for a near-neutral pH condition.

4. Conclusions and engineering implications

A novel strategy using pyrite-activated sodium percarbonate (FeS₂/SPC) was developed to degrade an emerging groundwater contaminant TCP. TCP could be effectively removed by FeS₂/SPC system with a much higher removal efficiency than classic Fenton system. Free radical quenching test, hydroxyl radical probe reaction, EPR, XPS, XRD and SEM were used together to elucidate the reaction mechanism. SPC was decomposed to produce H₂O₂, which was activated by Fe²⁺ released from pyrite to produce HO· and O₂^{·-}. Both HO· and O₂^{·-} contributed to TCP oxidation, but HO· played a more important role. The S elements of pyrite reduced Fe³⁺ to Fe²⁺, thus promoting the regeneration of Fe²⁺ and further production of HO· and O₂^{·-}. The Fe²⁺ regeneration enhanced the TCP removal efficiency. The removal efficiency of TCP was significantly enhanced with increases in concentrations of pyrite and SPC and decreases in pyrite particle size and pH. Although matrix species in real groundwater (e.g., HCO₃⁻, Cl⁻ and NOM) inhibited FeS₂/SPC oxidation of TCP, the reaction system was still capable of removing TCP.

Based on the above scientific discoveries, pyrite-activate SPC is a promising groundwater remediation technology. This innovative technology has three advantages. (1) The use of natural pyrite avoids injection of external activators into the aquifer, which reduces disturbance to the groundwater. (2) The S elements of pyrite promoted the regeneration of Fe²⁺, thus enhancing the TCP degradation. (3) SPC is more stable than liquid H₂O₂ and is safer to handle and transport.

CRediT authorship contribution statement

Chunyun Gu: Conceptualization, Investigation, Formal analysis, Visualization, Writing – original draft. **Shuyu Liu:** Data curation, Supervision. **Jiarui Liang:** Investigation, Supervision. **Yuqian Wang:** Investigation, Supervision. **Shuyu Lu:** Supervision. **Jie Ma:** Project administration, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

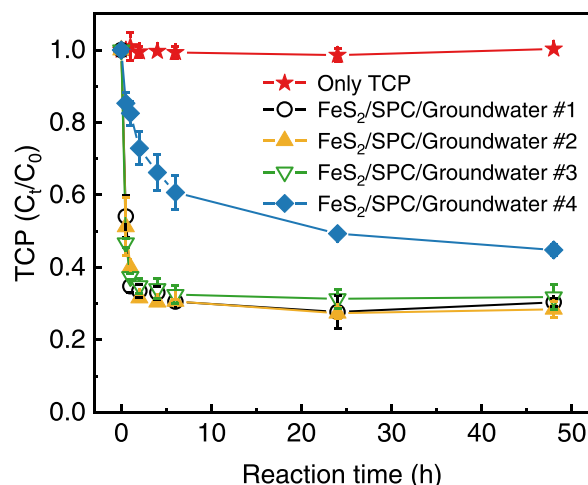


Fig. 8. Pyrite activated SPC for TCP degradation in different groundwater. Experimental condition: [TCP]₀ = 0.1 mM; [SPC]₀ = 1 mM; [FeS₂]₀ = 5 g/L; initial pH = 3.

Data Availability

Data will be made available on request.

Acknowledgements

The authors thank the financial support from National Natural Science Foundation of China (21878332, 42177042) and Science Foundation of China University of Petroleum-Beijing (2462022QNXZ006, 2462022YXZZ011).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2022.109217.

References

- [1] P. Dvorak, S. Bidmanova, J. Damborsky, Z. Prokop, Immobilized synthetic pathway for biodegradation of toxic recalcitrant pollutant 1,2,3-trichloropropane, *Environ. Sci. Technol.* 48 (2014) 6859–6866, <https://doi.org/10.1021/es500396r>.
- [2] J. Yan, B.A. Rash, F.A. Rainey, W.M. Moe, Isolation of novel bacteria within the Chloroflexi capable of reductive dechlorination of 1,2,3-trichloropropane, *Environ. Microbiol.* 11 (2009) 833–843, <https://doi.org/10.1111/j.1462-2920.2008.01804.x>.
- [3] K.R. Burow, W.D. Floyd, M.K. Landon, Factors affecting 1,2,3-trichloropropane contamination in groundwater in California, *Sci. Total Environ.* 672 (2019) 324–334, <https://doi.org/10.1016/j.scitotenv.2019.03.420>.
- [4] ECHA, 2012. Candidate list of substances of very high concern for authorization. European Chemicals Agency: Helsinki. <https://echa.europa.eu/candidate-list-table/-/dislist/details/0b0236e1807da1bc>.
- [5] USEPA, 2009. Toxicological Review of 1,2,3-Trichloropropane. U.S. Environmental Protection Agency: Washington, DC. <http://www.epa.gov/iris/toxreviews/0200tr.pdf>.
- [6] USEPA, 2017. Technical Fact Sheet-1,2,3-Trichloropropane (TCP). U.S. Environmental Protection Agency: Washington, DC. <https://www.epa.gov/fed/fac/technical-fact-sheet-123-trichloropropane-tcp>.
- [7] USEPA, 2012. The Third Unregulated Contaminant Monitoring Rule (UCMR 3). U. S. Environmental Protection Agency: Washington, DC. <https://www.epa.gov/dwu/cmr/third-unregulated-contaminant-monitoring-rule>.
- [8] L. Love, 1,2,3-Trichloropropane: California's Response to a Persistent Pollutant, *Am. Water Works Assoc.* 111 (2019) 34–43, <https://doi.org/10.1002/awwa.1411>.
- [9] R.G. Tardiff, M.L. Carson, Derivation of a reference dose and drinking water equivalent level for 1,2,3-trichloropropane, *Food Chem. Toxicol.* 48 (2010) 1488–1510, <https://doi.org/10.1016/j.fct.2010.03.016>.
- [10] M. Leistra, J. Boesten, Pesticide contamination of groundwater in western Europe, *Agric. Ecosyst. Environ.* 26 (1986) 369–389, [https://doi.org/10.1016/0167-8809\(89\)90018-2](https://doi.org/10.1016/0167-8809(89)90018-2).
- [11] H. Li, Z. Han, Y. Qian, X. Kong, P. Wang, In Situ persulfate oxidation of 1,2,3-trichloropropane in groundwater of North China Plain, *Int. J. Environ. Res. Public Health* 16 (2019) 1–16, <https://doi.org/10.3390/ijerph16152752>.

- [12] Tratnyek, P.G., Sarathy, V., Salter, A.J., Nurmi, J.T., Johnson, G.O., DeVoe, T., Lee, P., Prospects for Remediation of 1,2,3-Trichloropropane by Natural and Engineered Abiotic Degradation Reactions. Strategic Environmental Research and Development Program (SERDP): 2010; pp Project ER-1457.
- [13] M. Schmitt, S. Varadhan, S. Dworatzek, J. Webb, E. Suchomel, Optimization and validation of enhanced biological reduction of 1,2,3-trichloropropane in groundwater, *Remediat. J.* 28 (2017) 17–25, <https://doi.org/10.1002/rem.21539>.
- [14] A.J. Salter-Blanc, P.G. Tratnyek, Effects of solution chemistry on the dechlorination of 1,2,3-trichloropropane by zero-valent zinc, *Environ. Sci. Technol.* 45 (2011) 4073–4079, <https://doi.org/10.1021/es104081p>.
- [15] K.H. Wei, J. Ma, B.D. Xi, M.D. Yu, J. Cui, B.L. Chen, Y. Li, Q.B. Gu, X.S. He, Recent progress on in-situ chemical oxidation for the remediation of petroleum contaminated soil and groundwater, *J. Hazard Mater.* 432 (2022), 128738, <https://doi.org/10.1016/j.jhazmat.2022.128738>.
- [16] Y.S. Jung, W.T. Lim, J.Y. Park, Y.H. Kim, Effect of pH on Fenton and Fenton-like oxidation, *Environ. Technol.* 30 (2009) 183–190, <https://doi.org/10.1080/09593330802468848>.
- [17] A. Fischbacher, C. von Sonntag, T.C. Schmidt, Hydroxyl radical yields in the Fenton process under various pH, ligand concentrations and hydrogen peroxide/Fe(II) ratios, *Chemosphere* 182 (2017) 738–744, <https://doi.org/10.1016/j.chemosphere.2017.05.039>.
- [18] W. Jiang, P. Tang, S. Lu, Y. Xue, X. Zhang, Z. Qiu, Q. Sui, Comparative studies of H₂O₂/Fe(II)/formic acid, sodium percarbonate/Fe(II)/formic acid and calcium peroxide/Fe(II)/formic acid processes for degradation performance of carbon tetrachloride, *Chem. Eng. J.* 344 (2018) 453–461, <https://doi.org/10.1016/j.cej.2018.03.092>.
- [19] J. Huang, Z. Zhou, M. Ali, X. Gu, M. Danish, Q. Sui, S. Lyu, Degradation of trichloroethene by citric acid chelated Fe(II) catalyzing sodium percarbonate in the environment of sodium dodecyl sulfate aqueous solution, *Chemosphere* 281 (2021), 130798, <https://doi.org/10.1016/j.chemosphere.2021.130798>.
- [20] H. Che, S. Bae, W. Lee, Degradation of trichloroethylene by Fenton reaction in pyrite suspension, *J. Hazard. Mater.* 185 (2011) 1355–1361, <https://doi.org/10.1016/j.jhazmat.2010.10.055>.
- [21] Y. Lyu, S. Lyu, P. Tang, W. Jiang, Y. Sun, M. Li, Q. Sui, Degradation of trichloroethylene in aqueous solution by sodium percarbonate activated with Fe (II)-citric acid complex in the presence of surfactant Tween-80, *Chemosphere* 257 (2020), 122723, <https://doi.org/10.1016/j.chemosphere.2020.122723>.
- [22] S. Yu, X. Gu, S. Lu, Y. Xue, X. Zhang, M. Xu, Z. Qiu, Q. Sui, Degradation of phenanthrene in aqueous solution by a persulfate/percarbonate system activated with CA chelated-Fe(II), *Chem. Eng. J.* 333 (2018) 122–131, <https://doi.org/10.1016/j.cej.2017.09.158>.
- [23] Y. Zhou, M. Huang, X. Wang, J. Gao, G. Fang, D. Zhou, Efficient transformation of diethyl phthalate using calcium peroxide activated by pyrite, *Chemosphere* 253 (2020), 126662, <https://doi.org/10.1016/j.chemosphere.2020.126662>.
- [24] R. Matta, K. Hanna, S. Chiron, Fenton-like oxidation of 2,4,6-trinitrotoluene using different iron minerals, *Sci. Total Environ.* 385 (2007) 242–251, <https://doi.org/10.1016/j.scitotenv.2007.06.030>.
- [25] Z. Zhao, S. Pan, Y. Ye, X. Zhang, B. Pan, FeS₂/H₂O₂ mediated water decontamination from p-arsanilic acid via coupling oxidation, adsorption and coagulation: Performance and mechanism, *Chem. Eng. J.* 381 (2020), <https://doi.org/10.1016/j.cej.2019.122667>.
- [26] Q. Chen, Y. Yao, Z. Zhao, J. Zhou, Z. Chen, Long term catalytic activity of pyrite in heterogeneous fenton-like oxidation for the tertiary treatment of dyeing wastewater, *J. Environ. Chem. Eng.* 9 (2021), <https://doi.org/10.1016/j.jece.2021.105730>.
- [27] J.G. Kim, H.B. Kim, W.G. Jeong, K. Baek, Enhanced-oxidation of sulfanilamide in groundwater using combination of calcium peroxide and pyrite, *J. Hazard. Mater.* 419 (2021), 126514, <https://doi.org/10.1016/j.jhazmat.2021.126514>.
- [28] Z.H. Diao, W. Chu, FeS₂ assisted degradation of atrazine by bentonite-supported nZVI coupling with hydrogen peroxide process in water: Performance and mechanism, *Sci. Total Environ.* 754 (2021), 142155, <https://doi.org/10.1016/j.scitotenv.2020.142155>.
- [29] Z.H. Diao, X.R. Xu, D. Jiang, G. Li, J.J. Liu, L.J. Kong, L.Z. Zuo, Enhanced catalytic degradation of ciprofloxacin with FeS₂/SiO₂ microspheres as heterogeneous Fenton catalyst: kinetics, reaction pathways and mechanism, *J. Hazard Mater.* 327 (2017) 108–115, <https://doi.org/10.1016/j.jhazmat.2016.12.045>.
- [30] G.J. He, D.J. Zhong, Y.L. Xu, P. Liu, S.J. Zeng, S. Wang, Pyrite/H₂O₂/hydroxylamine system for efficient decolorization of rhodamine B, *Water Sci. Technol.* 83 (2021) 2218–2231, <https://doi.org/10.2166/wst.2021.135>.
- [31] L. Lefticariu, A. Schimmelmman, L.M. Pratt, E.M. Ripley, Oxygen isotope partitioning during oxidation of pyrite by H₂O₂ and its dependence on temperature, *Geochim. Et. Cosmochim. Acta* 71 (2007) 5072–5088, <https://doi.org/10.1016/j.gca.2007.08.022>.
- [32] T. Liu, N. Chen, Y. Deng, F. Chen, C. Feng, Degradation of p-nitrophenol by nanopyrite catalyzed Fenton reaction with enhanced peroxide utilization, *RSC Adv.* 10 (2020) 15901–15912, <https://doi.org/10.1039/d0ra01177k>.
- [33] W. Liu, Y. Wang, Z. Ai, L. Zhang, Hydrothermal synthesis of FeS₂ as a high-efficiency fenton reagent to degrade alachlor via superoxide-mediated Fe(II)/Fe (III) cycle, *ACS Appl. Mater. Interfaces* 7 (2015) 28534–28544, <https://doi.org/10.1021/acsami.5b09919>.
- [34] Z. Miao, X. Gu, S. Lu, M.L. Brusseau, N. Yan, Z. Qiu, Q. Sui, Enhancement effects of reducing agents on the degradation of tetrachloroethene in the Fe(II)/Fe(III) catalyzed percarbonate system, *J. Hazard. Mater.* 300 (2015) 530–537, <https://doi.org/10.1016/j.jhazmat.2015.07.047>.
- [35] X. Fu, X. Gu, S. Lu, Z. Miao, M. Xu, X. Zhang, M. Danish, H. Cui, U. Farooq, Z. Qiu, Q. Sui, Enhanced degradation of benzene by percarbonate activated with Fe(II)-glutamate complex, *Environ. Sci. Pollut. Res.* 23 (2015) 6758–6766, <https://doi.org/10.1007/s11356-015-5908-2>.
- [36] Y. Li, Y. Zhu, D. Wang, G. Yang, L. Pan, Q. Wang, B.J. Ni, H. Li, X. Yuan, L. Jiang, W. Tang, Fe(II) catalyzing sodium percarbonate facilitates the dewaterability of waste activated sludge: performance, mechanism, and implication, *Water Res.* 174 (2020), 115626, <https://doi.org/10.1016/j.watres.2020.115626>.
- [37] H. Cui, X. Gu, S. Lu, X. Fu, X. Zhang, G.Y. Fu, Z. Qiu, Q. Sui, Degradation of ethylbenzene in aqueous solution by sodium percarbonate activated with EDDS-Fe (III) complex, *Chem. Eng. J.* 309 (2017) 80–88, <https://doi.org/10.1016/j.cej.2016.10.029>.
- [38] X. Fu, X. Gu, S. Lu, V.K. Sharma, M.L. Brusseau, Y. Xue, M. Danish, G.Y. Fu, Z. Qiu, Q. Sui, Benzene oxidation by Fe(III)-activated percarbonate: matrix-constituent effects and degradation pathways, *Chem. Eng. J.* 309 (2016) 22–29, <https://doi.org/10.1016/j.cej.2016.10.006>.
- [39] X. Liu, S. Yuan, M. Tong, D. Liu, Oxidation of trichloroethylene by the hydroxyl radicals produced from oxygenation of reduced nontronite, *Water Res.* 113 (2017) 72–79, <https://doi.org/10.1016/j.watres.2017.02.012>.
- [40] G.J. He, D.J. Zhong, Y.L. Xu, P. Liu, S.J. Zeng, S. Wang, Pyrite/H₂O₂/hydroxylamine system for efficient decolorization of rhodamine B, *Water Sci. Technol.* 83 (2021) 2218–2231, <https://doi.org/10.2166/wst.2021.135>.
- [41] M. Cai, J. Yao, H. Yang, R. Wang, K. Masakoral, Aerobic biodegradation process of petroleum and pathway of main compounds in water flooding well of Dagang oil field, *Bioresour. Technol.* 144 (2013) 100–106, <https://doi.org/10.1016/j.biortech.2013.06.082>.
- [42] E.J. Kim, J.H. Kim, Y.S. Chang, D. Turcio-Ortega, P.G. Tratnyek, Effects of metal ions on the reactivity and corrosion electrochemistry of Fe/FeS nanoparticles, *Environ. Sci. Technol.* 48 (2014) 4002–4011, <https://doi.org/10.1021/es405622d>.
- [43] W. Lee, B. Batchelor, Abiotic reductive dechlorination of chlorinated ethylenes by iron-bearing soil minerals. 2, *Green Rust. Environ. Sci. Technol.* 36 (2002) 5348–5354, <https://doi.org/10.1021/es0258374>.
- [44] D. Cheng, S. Yuan, P. Liao, P. Zhang, Oxidizing impact induced by mackinawite (FeS) nanoparticles at oxic conditions due to production of hydroxyl radicals, *Environ. Sci. Technol.* 50 (2016) 11646–11653, <https://doi.org/10.1021/acs.est.6b02833>.
- [45] H. Chen, Z. Zhang, Z. Yang, Q. Yang, B. Li, Z. Bai, Heterogeneous fenton-like catalytic degradation of 2,4-dichlorophenoxyacetic acid in water with FeS, *Chem. Eng. J.* 273 (2015) 481–489, <https://doi.org/10.1016/j.cej.2015.03.079>.
- [46] E.J. Kim, K. Murugesan, J.H. Kim, P.G. Tratnyek, Y.S. Chang, Remediation of trichloroethylene by FeS-coated iron nanoparticles in simulated and real groundwater: effects of water chemistry, *Ind. Eng. Chem. Res.* 52 (2013) 9343–9350, <https://doi.org/10.1021/ie400165a>.
- [47] S. Tang, Z. Wang, D. Yuan, C. Zhang, Y. Rao, Z. Wang, K. Yin, Ferrous ion-tartaric acid chelation promoted calcium peroxide fenton-like reactions for simulated organic wastewater treatment, *J. Clean. Prod.* 268 (2020), 122253, <https://doi.org/10.1016/j.jclepro.2020.122253>.
- [48] Y. Zhu, J. Qin, S. Zhang, A. Radian, M. Long, Solid peroxides in Fenton-like reactions at near neutral pHs: Superior performance of MgO₂ on the accelerated reduction of ferric species, *Chemosphere* 270 (2021), 128639, <https://doi.org/10.1016/j.chemosphere.2020.128639>.
- [49] X. Fu, M.L. Brusseau, X. Zang, S. Lu, X. Zhang, U. Farooq, Z. Qiu, Q. Sui, Enhanced effect of HAH on citric acid-chelated Fe(II)-catalyzed percarbonate for trichloroethene degradation, *Environ. Sci. Pollut. Res.* 24 (2017) 24318–24326, <https://doi.org/10.1007/s11356-017-0070-7>.
- [50] G.V. Buxton, C.L. Greenstock, W.P. Helman, A.B. Ross, Critical review of rate constants for reactions of hydrated electrons, hydrogen atoms and hydroxyl radicals (•OH/•O₂) in aqueous solution, *J. Phys. Chem. Ref. Data.* 17 (1988) 513–886, <https://doi.org/10.1063/1.555805>.
- [51] M. Ali, X. Zhang, A. Idrees, M. Tariq, M. Danish, U. Farooq, A. Shan, X. Jiang, J. Huang, S. Lyu, Advancement in Fenton-like reactions using PVA coated calcium peroxide/FeS system: pivotal role of sulfide ion in regenerating the Fe(II) ions and improving trichloroethylene degradation, *J. Environ. Chem. Eng.* 9 (2021), <https://doi.org/10.1016/j.jece.2020.104591>.
- [52] U. Farooq, M. Danish, S. Lu, M. Naqvi, X. Gu, X. Fu, X. Zhang, M. Nasir, Synthesis of nZVI@reduced graphene oxide: an efficient catalyst for degradation of 1,1,1-trichloroethane (TCA) in percarbonate system, *Res. Chem. Intermed.* 43 (2016) 3219–3236, <https://doi.org/10.1007/s11164-016-2821-3>.
- [53] Y.Y. Jiang, Z.W. Chen, M.M. Li, Q.H. Xiang, X.X. Wang, H.F. Miao, W.Q. Ruan, Degradation of diclofenac sodium using Fenton-like technology based on nano-calcium peroxide, *Sci. Total Environ.* 773 (2021), 144801, <https://doi.org/10.1016/j.scitotenv.2020.144801>.
- [54] M. Zhao, Y. Xiang, X. Jiao, B. Cao, S. Tang, Z. Zheng, X. Zhang, T. Jiao, D. Yuan, MoS₂ co-catalysis promoted CaO₂ Fenton-like process: performance and mechanism, *Sep. Purif. Technol.* (2021) 276, <https://doi.org/10.1016/j.seppur.2021.119289>.
- [55] J.J. Wu, M. Muruganandham, J.S. Yang, S.S. Lin, Oxidation of DMSO on goethite catalyst in the presence of H₂O₂ at neutral pH, *Catal. Commun.* 7 (2006) 901–906, <https://doi.org/10.1016/j.catcom.2006.03.015>.
- [56] F. Sun, T. Chen, X. Zou, H. Liu, Z. Chu, D. Shu, H. Wang, F. Huang, D. Chen, A quantitative analysis of hydroxyl radical generation as H₂O₂ encounters siderite: Kinetics and effect of parameters, *Appl. Geochem.* 126 (2021), <https://doi.org/10.1016/j.apgeochem.2021.104893>.
- [57] D. Cheng, A. Neumann, S. Yuan, W. Liao, A. Qian, Oxidative degradation of organic contaminants by FeS in the presence of O₂, *Environ. Sci. Technol.* 54 (2020) 4091–4101, <https://doi.org/10.1021/acs.est.9b07012>.

- [58] N. Yan, F. Liu, W. Huang, Interaction of oxidants in siderite catalyzed hydrogen peroxide and persulfate system using trichloroethylene as a target contaminant, *Chem. Eng. J.* 219 (2013) 149–154, <https://doi.org/10.1016/j.cej.2012.12.072>.
- [59] Y. Zhang, H.P. Tran, I. Hussain, Y. Zhong, S. Huang, Degradation of p-chloroaniline by pyrite in aqueous solutions, *Chem. Eng. J.* 279 (2015) 396–401, <https://doi.org/10.1016/j.cej.2015.03.016>.
- [60] J.D. Laats, T.G. Le, Kinetics and modeling of the Fe(III)/H₂O₂ system in the presence of sulfate in acidic aqueous solutions, *Environ. Sci. Technol.* 39 (2005) 1811–1818, <https://doi.org/10.1021/es0493648>.
- [61] Q. Wang, A.T. Lemley, Kinetic effect of humic acid on alachlor degradation by anodic fenton treatment, *J. Environ. Qual.* 33 (2004) 2343–2352, <https://doi.org/10.2134/jeq2004.2343>.
- [62] X. Zhu, X. Qian, *Groundwater Hydrology*, China Environmental Science Press, Beijing, 2005, pp. 98–99.
- [63] C. Fan, L. Tsui, M.C. Liao, Parathion degradation and its intermediate formation by Fenton process in neutral environment, *Chemosphere* 82 (2011) 229–236, <https://doi.org/10.1016/j.chemosphere.2010.10.016>.